

Statistics & Probability

ANSWER KEY



Permutations

- 1 How many ways can 5 distinct books be arranged on a shelf?

$$5! = 120.$$

- 2 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?

$${}_{12}P_3 = 12 \times 11 \times 10 = 1320.$$

- 3 How many distinct arrangements are there of the letters in the word APPLE?

There are 5 letters with the letter P repeated twice: $\frac{5!}{2!} = \frac{120}{2} = 60$.

Combinations

- 4 How many ways can a committee of 4 be chosen from 11 people?

$$\binom{11}{4} = \frac{11!}{4! 7!} = 330.$$

- 5 A pizza shop offers 8 toppings. How many different 3-topping pizzas can be made?

$$\binom{8}{3} = \frac{8 \times 7 \times 6}{6} = 56.$$

- 6 From a group of 6 men and 5 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?

$$\binom{6}{2} \binom{5}{2} = 15 \times 10 = 150.$$

Probability rules

- 7 A fair die is rolled. Find $P(\text{even or greater than 4})$.

Even = {2, 4, 6}, greater than 4 = {5, 6}. Union = {2, 4, 5, 6}. $P = \frac{4}{6} = \frac{2}{3}$.

- 8 $P(A) = 0.45$ and $P(B) = 0.3$, and A and B are mutually exclusive. Find $P(A \text{ or } B)$.

$$P(A \text{ or } B) = 0.45 + 0.3 = 0.75.$$

- 9 A card is drawn from a standard 52-card deck. Find $P(\text{not a face card})$.

There are 12 face cards, so 40 are not face cards. $P = \frac{40}{52} = \frac{10}{13}$.

Conditional probability and independence

- 10 $P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Are A and B independent?

$P(A)P(B) = 0.5 \times 0.4 = 0.2 = P(A \cap B)$. Yes, A and B are independent.

- 11 In a class, 60% of students play sports, and 25% of all students both play sports and are honor students. Find $P(\text{honor student} \mid \text{plays sports})$.

$$P(\text{honor} \mid \text{sports}) = \frac{0.25}{0.6} \approx 0.417, \text{ or about } 41.7\%.$$

- 12 Events C and D are independent, with $P(C) = 0.7$ and $P(D) = 0.6$. Find $P(C \cup D)$.

$$P(C \cap D) = 0.7 \times 0.6 = 0.42. P(C \cup D) = 0.7 + 0.6 - 0.42 = 0.88.$$

Measures of center and spread

- 13 Find the mean and median of the data set 12, 15, 18, 20, 25.

$$\text{Mean} = \frac{12 + 15 + 18 + 20 + 25}{5} = \frac{90}{5} = 18. \text{ Median (middle value, already sorted)} = 18.$$

- 14 Find the population standard deviation of the data set 4, 8, 6, 5, 3.

$$\text{Mean} = \frac{26}{5} = 5.2. \text{ Squared deviations: } 1.44, 7.84, 0.64, 0.04, 4.84, \text{ summing to } 14.8. \text{ Variance} = \frac{14.8}{5} = 2.96. \text{ Standard deviation} = \sqrt{2.96} \approx 1.72.$$

Data representation

- 15 The bar chart shows the distribution of grades in a class of 37 students. Find the probability that a randomly selected student earned a grade of A or B, as a percent to the nearest whole percent.

$$P(A \text{ or } B) = \frac{8 + 15}{37} = \frac{23}{37} \approx 0.622, \text{ or about } 62\%.$$

- 16 Using the same bar chart, find the probability that a randomly selected student scored a C or D grade, as a percent to the nearest whole percent.

$$P(C \text{ or } D) = \frac{10 + 4}{37} = \frac{14}{37} \approx 0.378, \text{ or about } 38\%.$$

The binomial theorem

- 17 Find the coefficient of x^3y^2 in the expansion of $(x + y)^5$.

The term is $\binom{5}{2}x^3y^2$, so the coefficient is $\binom{5}{2} = 10$.

- 18 In the expansion of $(2x - 1)^4$, find the coefficient of x^2 .

The general term is $\binom{4}{k}(2x)^{4-k}(-1)^k$. For x^2 , $4 - k = 2 \Rightarrow k = 2$: $\binom{4}{2}(2x)^2(-1)^2 = 6(4x^2) = 24x^2$. The coefficient is 24.

The normal distribution

- 19 A data set is normally distributed with mean 100 and standard deviation 15. What percent of values lie between 70 and 130?

70 and 130 are each 2 standard deviations from the mean (100 ± 30). By the 68-95-99.7 rule, about 95% of values lie in this range.

- 20 For the same distribution (mean 100, standard deviation 15), find the approximate percentage of values greater than 115.

115 is 1 standard deviation above the mean. About 68% of values lie within 1 standard deviation, so 34% lie between the mean and 115. Since 50% lie above the mean, the percentage above 115 is $50\% - 34\% = 16\%$.

A well-designed word problem: quality control sampling

- 21 This question has three parts. A factory produces batteries, and 4% of batteries are defective. A quality-control inspector randomly selects a sample of 3 batteries from a large shipment (assume the selections are independent). (a) Find the probability that none of the 3 batteries are defective. (b) Find the probability that exactly 1 of the 3 batteries is defective. (c) Find the probability that at least 1 of the 3 batteries is defective.

(a) $P(\text{none defective}) = (0.96)^3 = 0.884736 \approx 88.5\%$. (b)
 $P(\text{exactly 1}) = \binom{3}{1}(0.04)(0.96)^2 = 3(0.04)(0.9216) = 0.110592 \approx 11.1\%$. (c)
 $P(\text{at least 1}) = 1 - P(\text{none}) = 1 - 0.884736 = 0.115264 \approx 11.5\%$.

Multiple choice — easy

- 22 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 8 members?

- a. 512
- b. 330
- c. 342
- d. 336 ✓

$${}_8P_3 = 8 \times 7 \times 6 = 336.$$

- 23 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 18 members?

- a. 4902
- b. 4890
- c. 4896 ✓
- d. 5832

$${}_{18}P_3 = 18 \times 17 \times 16 = 4896.$$

- 24 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 16 members?

- a. 3360 ✓
- b. 4096
- c. 3366
- d. 3354

$${}_{16}P_3 = 16 \times 15 \times 14 = 3360.$$

- 25 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 13 members?

- a. 2197
- b. 1722
- c. 1716 ✓
- d. 1710

$${}_{13}P_3 = 13 \times 12 \times 11 = 1716.$$

- 26 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 10 members?

- a. 714
- b. 726
- c. 720 ✓
- d. 1000

$${}_{10}P_3 = 10 \times 9 \times 8 = 720.$$

- 27 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 14 members?

- a. 2178
- b. 2184 ✓
- c. 2190
- d. 2744

$${}_{14}P_3 = 14 \times 13 \times 12 = 2184.$$

28 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 12 members?

a. 1326

b. 1314

c. 1320 ✓

d. 1728

$${}_{12}P_3 = 12 \times 11 \times 10 = 1320.$$

29 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 20 members?

a. 6834

b. 6840 ✓

c. 6846

d. 8000

$${}_{20}P_3 = 20 \times 19 \times 18 = 6840.$$

30 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 17 members?

a. 4913

b. 4080 ✓

c. 4086

d. 4074

$${}_{17}P_3 = 17 \times 16 \times 15 = 4080.$$

- 31** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 9 members?

a. 504 ✓

b. 498

c. 729

d. 510

$${}_9P_3 = 9 \times 8 \times 7 = 504.$$

- 32** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 19 members?

a. 6859

b. 5808

c. 5814 ✓

d. 5820

$${}_{19}P_3 = 19 \times 18 \times 17 = 5814.$$

- 33** In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 7 members?

a. 343

b. 210 ✓

c. 204

d. 216

$${}_7P_3 = 7 \times 6 \times 5 = 210.$$

- 34 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 6 members?

a. 216
b. 114
c. 120 ✓
d. 126

$${}_6P_3 = 6 \times 5 \times 4 = 120.$$

- 35 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 15 members?

a. 2736
b. 3375
c. 2730 ✓
d. 2724

$${}_{15}P_3 = 15 \times 14 \times 13 = 2730.$$

- 36 In how many ways can a president, vice-president, and secretary (three distinct roles) be chosen from a club of 11 members?

a. 996
b. 984
c. 1331
d. 990 ✓

$${}_{11}P_3 = 11 \times 10 \times 9 = 990.$$

Multiple choice — medium

37 How many ways can a committee of 6 be chosen from 11 people?

a. 330

b. 462 ✓

c. 465

d. 459

$$\binom{11}{6} = 462.$$

38 How many ways can a committee of 5 be chosen from 11 people?

a. 465

b. 459

c. 462 ✓

d. 467

$$\binom{11}{5} = 462.$$

39 How many ways can a committee of 4 be chosen from 15 people?

a. 1365 ✓

b. 1362

c. 3003

d. 1368

$$\binom{15}{4} = 1365.$$

40 How many ways can a committee of 5 be chosen from 19 people?

a. 11631

b. 27132

c. 11628 ✓

d. 11625

$$\binom{19}{5} = 11628.$$

41 How many ways can a committee of 4 be chosen from 14 people?

a. 1004

b. 1001 ✓

c. 998

d. 2002

$$\binom{14}{4} = 1001.$$

42 How many ways can a committee of 5 be chosen from 7 people?

a. 21 ✓

b. 24

c. 18

d. 7

$$\binom{7}{5} = 21.$$

43 How many ways can a committee of 4 be chosen from 17 people?

a. 2383

b. 6188

c. 2380 ✓

d. 2377

$$\binom{17}{4} = 2380.$$

44 How many ways can a committee of 3 be chosen from 10 people?

a. 117

b. 123

c. 210

d. 120 ✓

$$\binom{10}{3} = 120.$$

45 How many ways can a committee of 4 be chosen from 20 people?

a. 4842

b. 4845 ✓

c. 4848

d. 15504

$$\binom{20}{4} = 4845.$$

46 How many ways can a committee of 3 be chosen from 6 people?

a. 20 ✓

b. 23

c. 17

d. 15

$$\binom{6}{3} = 20.$$

47 How many ways can a committee of 5 be chosen from 16 people?

a. 8008

b. 4371

c. 4365

d. 4368 ✓

$$\binom{16}{5} = 4368.$$

48 How many ways can a committee of 2 be chosen from 18 people?

a. 156

b. 150

c. 153 ✓

d. 816

$$\binom{18}{2} = 153.$$

49 How many ways can a committee of 4 be chosen from 7 people?

a. 21

b. 32

c. 38

d. 35 ✓

$$\binom{7}{4} = 35.$$

50 How many ways can a committee of 3 be chosen from 19 people?

a. 972

b. 969 ✓

c. 3876

d. 966

$$\binom{19}{3} = 969.$$

51 How many ways can a committee of 4 be chosen from 10 people?

a. 213

b. 252

c. 210 ✓

d. 207

$$\binom{10}{4} = 210.$$

Multiple choice — hard

52 Events C and D are independent, with $P(C) = 0.2$ and $P(D) = 0.9$. Find $P(C \cup D)$.

a. 0.97

b. 0.82

c. 0.92 ✓

d. 1.02

$$P(C \cap D) = 0.2 \times 0.9 = 0.18. P(C \cup D) = 0.2 + 0.9 - 0.18 = 0.92.$$

53 Events C and D are independent, with $P(C) = 0.4$ and $P(D) = 0.2$. Find $P(C \cup D)$.

a. 0.62

b. 0.52 ✓

c. 0.57

d. 0.42

$$P(C \cap D) = 0.4 \times 0.2 = 0.08. P(C \cup D) = 0.4 + 0.2 - 0.08 = 0.52.$$

54 Events C and D are independent, with $P(C) = 0.3$ and $P(D) = 0.3$. Find $P(C \cup D)$.

- a. 0.56
- b. 0.41
- c. 0.61
- d. 0.51 ✓

$$P(C \cap D) = 0.3 \times 0.3 = 0.09. P(C \cup D) = 0.3 + 0.3 - 0.09 = 0.51.$$

55 Events C and D are independent, with $P(C) = 0.2$ and $P(D) = 0.8$. Find $P(C \cup D)$.

- a. 0.74
- b. 0.84 ✓
- c. 0.89
- d. 0.94

$$P(C \cap D) = 0.2 \times 0.8 = 0.16. P(C \cup D) = 0.2 + 0.8 - 0.16 = 0.84.$$

56 Events C and D are independent, with $P(C) = 0.6$ and $P(D) = 0.6$. Find $P(C \cup D)$.

- a. 0.94
- b. 0.89
- c. 0.84 ✓
- d. 0.74

$$P(C \cap D) = 0.6 \times 0.6 = 0.36. P(C \cup D) = 0.6 + 0.6 - 0.36 = 0.84.$$

57 Find the coefficient of x^4y^6 in the expansion of $(x + y)^{10}$.

- a. 210 ✓
- b. 212
- c. 208
- d. 120

The term is $\binom{10}{4}x^4y^6$, so the coefficient is $\binom{10}{4} = 210$.

58 Find the coefficient of x^7y^3 in the expansion of $(x + y)^{10}$.

a. 118

b. 122

c. 120 ✓

d. 210

The term is $\binom{10}{7}x^7y^3$, so the coefficient is $\binom{10}{7} = 120$.

59 Find the coefficient of x^1y^5 in the expansion of $(x + y)^6$.

a. 6 ✓

b. 1

c. 4

d. 8

The term is $\binom{6}{1}x^1y^5$, so the coefficient is $\binom{6}{1} = 6$.

60 Find the coefficient of x^6y^4 in the expansion of $(x + y)^{10}$.

a. 210 ✓

b. 212

c. 252

d. 208

The term is $\binom{10}{6}x^6y^4$, so the coefficient is $\binom{10}{6} = 210$.

61 Find the coefficient of x^6y^3 in the expansion of $(x + y)^9$.

- a. 82
- b. 126
- c. 86
- d. 84 ✓

The term is $\binom{9}{6}x^6y^3$, so the coefficient is $\binom{9}{6} = 84$.

62 From a group of 10 men and 5 women, how many ways can a committee of 6 be formed with exactly 4 men and 2 women?

- a. 2110
- b. 2100 ✓
- c. 220
- d. 2090

$$\binom{10}{4}\binom{5}{2} = 210 \times 10 = 2100.$$

63 From a group of 10 men and 8 women, how many ways can a committee of 5 be formed with exactly 3 men and 2 women?

- a. 148
- b. 3350
- c. 3370
- d. 3360 ✓

$$\binom{10}{3}\binom{8}{2} = 120 \times 28 = 3360.$$

64 From a group of 10 men and 5 women, how many ways can a committee of 5 be formed with exactly 3 men and 2 women?

a. 1210

b. 130

c. 1190

d. 1200 ✓

$$\binom{10}{3}\binom{5}{2} = 120 \times 10 = 1200.$$

65 From a group of 7 men and 7 women, how many ways can a committee of 4 be formed with exactly 2 men and 2 women?

a. 42

b. 441 ✓

c. 431

d. 451

$$\binom{7}{2}\binom{7}{2} = 21 \times 21 = 441.$$

66 From a group of 10 men and 7 women, how many ways can a committee of 7 be formed with exactly 4 men and 3 women?

a. 7340

b. 7360

c. 7350 ✓

d. 245

$$\binom{10}{4}\binom{7}{3} = 210 \times 35 = 7350.$$